Automatic Detection of Scoria Cones from DEMs

Gabriele Berruti (4791919)
Computational Vision — University of Genoa

1 Introduction

This project detects *scoria cones* from single-channel Digital Elevation Models (DEMs), where each pixel encodes a terrain elevation in metres. The core contribution is the **separation of nested or adjacent cones**: point-based blob detectors collapse two overlapping craters into a single response, and a one-to-one matching then discards one of the two. We address this with a watershed segmentation of the crater depressions, which natively splits adjacent catchment basins.

We work on three datasets: **El Hierro** (5 m resolution, with ground-truth annotations, used for development and validation), **Etna** (2 m, noisier ground truth, used as a stress test) and **Jeju** (10 m, no ground truth, used for qualitative and unsupervised analysis).

2 Data and preprocessing

2.1 Step 0 — DEM loading and inspection

A DEM is a *range image*: a single elevation band. Each dataset uses a different *nodata* sentinel, which we convert to `NaN` so that all later computations ignore invalid pixels naturally. Table 1 summarises the three DEMs after loading.

In Table 1, $H \times W$ is the raster size in pixels (matrix rows $\times$ columns), Mpx the total number of pixels in millions ($H \cdot W$, a proxy for the memory footprint), NaN% the share of pixels that were *nodata* and got mapped to `NaN` (terrain outside the raster/island, ignored by every later step), Res. the ground size of one pixel in metres, dtype the numeric type of the elevation values, and $z_{\max}$ the maximum elevation (sanity check against the real volcano).

Dataset	Size (H×W)	Mpx	NaN%	Res. (m)	dtype	$z_{\max}$ (m)
El Hierro	5081×5681	28.9	40%	5	float32	1499.8
Etna	27756×25717	713.8	25%	2	float32	3324.1
Jeju	4364×7761	33.9	27%	10	float32	1945.0

Table 1: DEM inspection. Maximum elevations match the real volcanoes (Etna ≈3.3 km, El Hierro ≈1.5 km, Hallasan/Jeju ≈1.95 km), confirming correct loading.

Three facts drive the following steps: (i) Etna is ≈714 Mpx (≈2.86 GB as float32), so it must be processed in *tiles* rather than loaded whole; (ii) the three *nodata* conventions differ (-3.4×10^{38}, -9999, `NaN`) and are all mapped to `NaN`; (iii) each dataset has its own UTM coordinate reference system, so any shapefile must be reprojected into the CRS of its DEM before use.

2.2 Step 0.5 — Visual inspection

Before deriving any channel we inspect the data on El Hierro. Figure 1 shows that the raw DEM looks visually flat, whereas the Eduard renderings (NE/NW shading and texture shading) reveal the cones; this motivates the derived channels of Step 1. Figure 2 shows the 196 crater outlines of the ground truth: the two closest craters lie only 25 m apart (5 pixels) and several cones are adjacent or nested. Point-based blob detectors merge such pairs into a single detection — separating them is the goal of the watershed step.

For clarity, the panels in Figure 1 are: the **raw DEM**, i.e. the original elevation grid where each pixel is a terrain height in metres (it looks flat because a cone is only tens of metres tall relative to the whole island); the **Eduard renderings**, relief-shaded views produced by the *Eduard* relief-shading tool and shipped with the dataset; the **NE/NW shading** (hillshade), where the terrain is lit by a simulated sun from the North-East/North-West so that slopes facing the light are bright and those facing away are dark (two opposite directions are given so no slope stays in shadow); and the **texture shading**, a light-direction-independent rendering that emphasises fine surface detail.

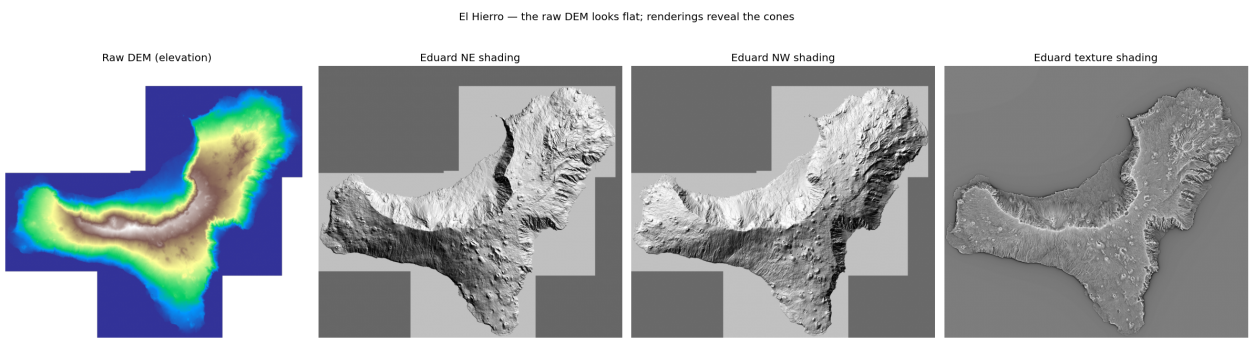

Figure 1: El Hierro: the raw DEM (left) is visually flat, while the Eduard renderings reveal the conical landforms.

Figure 3 isolates the cones that actually *overlap*, i.e. whose nearest neighbour is closer than the sum of the two radii. These are 8 craters in 4 pairs; note that they are not the single closest centroids of Figure 2 (two small craters), but pairs of *large* adjacent cones whose outlines intersect. These are the configurations a point-based detector merges into one and the watershed step must separate.

Ground-truth quality and label semantics. The El Hierro shapefile is *dirty* and mixes three feature types in a single file. The crater outlines (Id= 2) are open polylines: 191 of 196 do not close (median start–end gap 40 m, up to 556 m) and some have as few as 2 vertices. We therefore use crater *centroids* (plus an approximate radius) rather than the exact outline shape. A geometry test then characterises the three labels (Table 2): Id= 1 is the cone **base** (large, round, and enclosing the crater in 98% of cones), Id= 2 the **crater** rim, and Id= 3 a **fissure** — a short, strongly elongated segment that always carries a non-zero azimuth in the `Prj_Az` field. The `pericolosi` field is a hazard level (0/1/2). We adopt Id= 2 as the detection target.

Observation — what these data fix for the pipeline. The geometry tests turn the three labels into three decisions. (i) The **crater** (Id= 2) is the detection target: it is the only label present on *every* cone (196), and since its outlines are open we represent each crater by its **centroid and equivalent radius** rather than by the polygon. (ii) The **base** (Id= 1) encloses the crater in 98% of

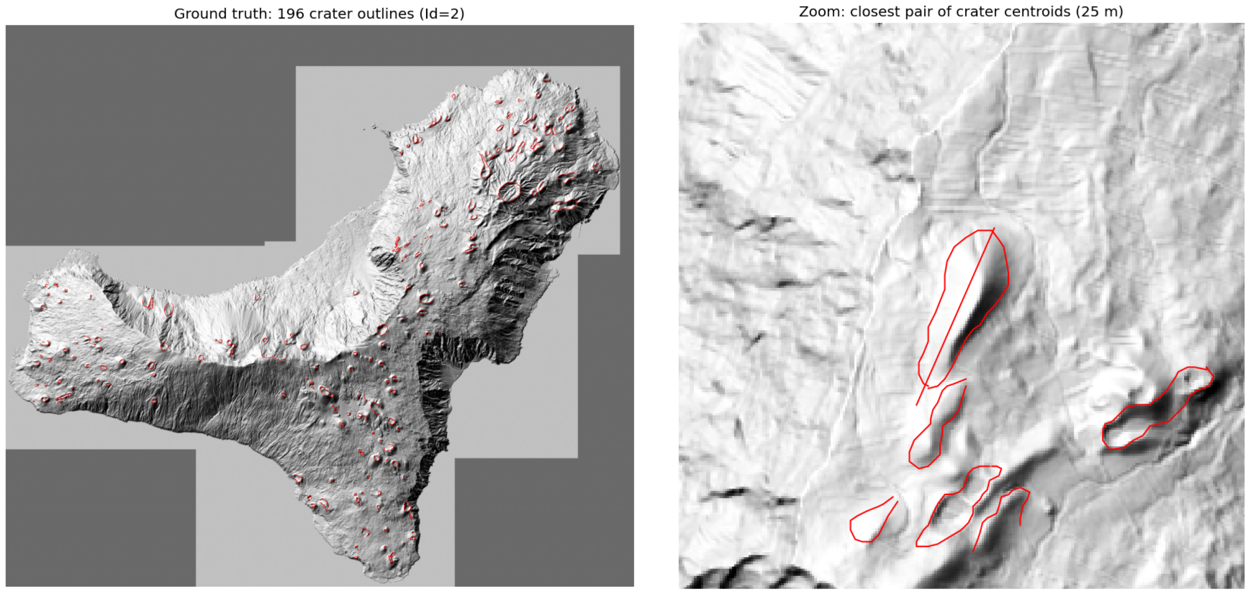

Figure 2: Ground truth on El Hierro. Left: the 196 crater outlines (Id= 2). Right: a zoom on the closest pair of crater centroids (25 m apart).

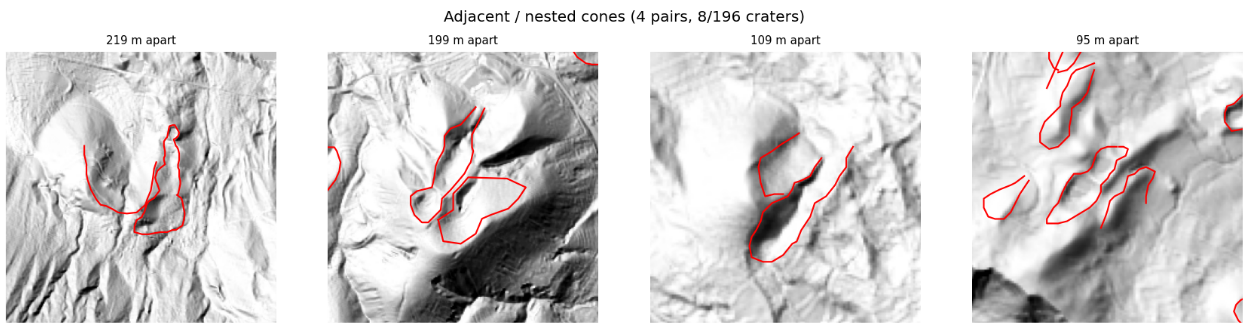

Figure 3: The adjacent/nested cases on El Hierro: the 4 crater pairs whose outlines overlap (8 craters). The watershed step is designed to split exactly these.

cones — this containment is exactly the nested configuration our contribution must preserve, and it gives a free sanity check on a detection. (iii) The **fissure** (Id= 3), a short segment with a real azimuth, is not a target but feeds the qualitative cone–fissure alignment of the bonus step. Finally, the labels are *not* equinumerous (151 base / 196 crater / 171 fissure): annotations are incomplete, so the whole pipeline is anchored on the complete crater set.

2.3 Step 0.6 — Quick data analysis

Three cheap measurements on El Hierro parameterise the later steps rather than guessing their constants. (i) The three Eduard renderings share the DEM grid exactly (same 5081×5681 shape and transform), so overlays are pixel-exact. (ii) The equivalent radius of the crater outlines (Figure 4, left) has median 71 m and a 5th–95th percentile range of 20–174 m, which we adopt as the LoG detector scale range. (iii) The nearest-neighbour distance between craters (Figure 4, right) has median 418 m but a minimum of 25 m, and 8/196 craters (4%) lie closer to a neighbour than the

Id	meaning	n	verts	length (m)	aspect	Prj_Az≠ 0
1	base	151	29	846	1.29	0%
2	crater	196	19	457	1.47	0%
3	fissure	171	2	272	8.32	100%

Table 2: Data-driven characterisation of the El Hierro labels (median values). The high aspect ratio and the always-present azimuth identify Id= 3 as fissures.

sum of the two radii, i.e. they are adjacent or nested. This 4% is precisely the population that point-based detection merges and that the watershed step is designed to recover.

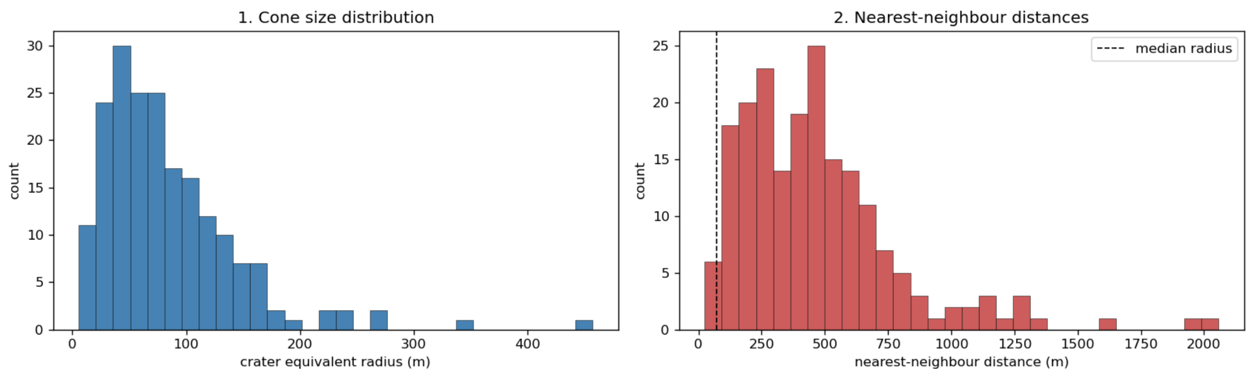

Figure 4: El Hierro crater statistics. Left: equivalent-radius distribution, used to set the LoG scale range. Right: nearest-neighbour distance distribution; the dashed line is the median radius, and the cones below it are the adjacent/nested cases.

Observation — what these three checks fix. Put together, the measurements set three things at once. The **alignment** lets every DEM-derived channel and the supplied shadings be stacked as co-registered inputs with no resampling; the crater **sizes** fix the LoG search range to 20–174 m (so the detector looks only at these radii); and the **adjacency** count shows that only 4% of cones overlap, yet this 4% is exactly the population a point-based detector merges into one and that the watershed step must recover. In short these numbers parameterise the detector (scale range), guarantee clean multi-channel inputs (alignment), and size the problem the contribution must solve (the 4% adjacent/nested cones).

3 Detection pipeline

3.1 Step 1 — Morphometric channels

From the single elevation band we derive three channels with the image gradient (deck `01_image_processing`, p.13): the **slope** $= \arctan\sqrt{g_x^2 + g_y^2}$ in degrees, the **aspect** $= \operatorname{atan2}(g_y, g_x)$ (the direction the slope faces), and the **roughness**, the local standard deviation of elevation in a 5×5 window. Figure 5 shows them on El Hierro. As a quantitative check, the median slope on the crater rims is 18.8° against 8.1° on the background (about 2.3× steeper), and the aspect rotates around each cone (bottom-right panel).

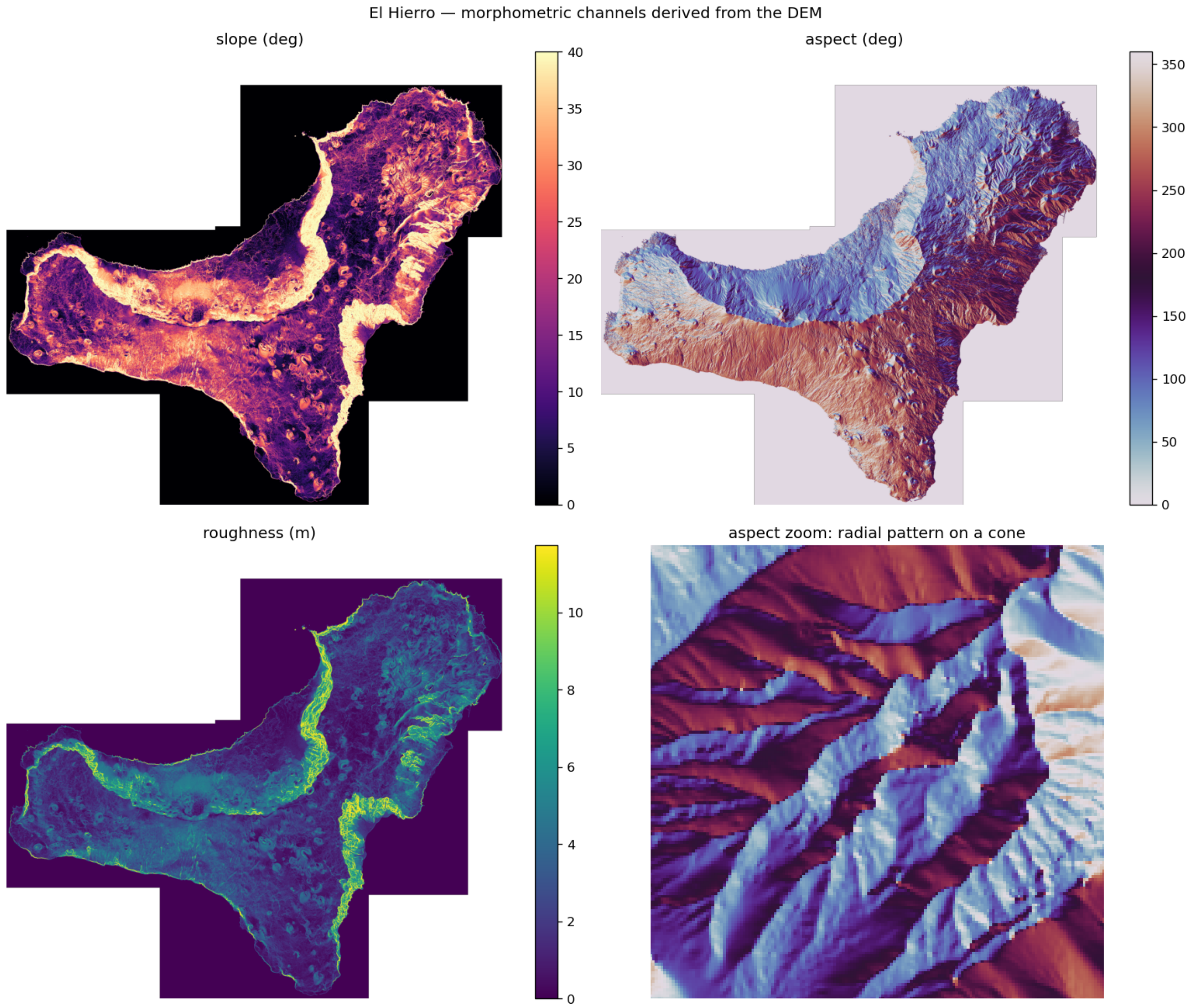

Figure 5: Morphometric channels derived from the El Hierro DEM: slope, aspect and roughness, plus a zoom of the aspect showing the radial pattern around a single cone.

Cones thus appear as bright *rings* in slope/roughness and as radial *colour wheels* in aspect; the only confounder is the coastal cliff, as steep as a crater rim, which is why Step 3 combines slope with roughness rather than thresholding slope alone. These three channels, together with the texture rendering, are the input of the detection steps.